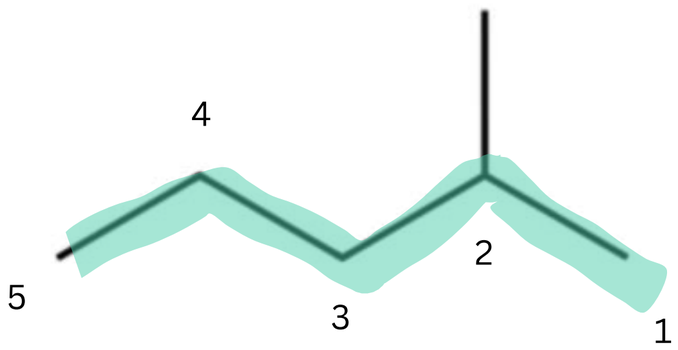
A01-FOUNDATION C_6H_{14}

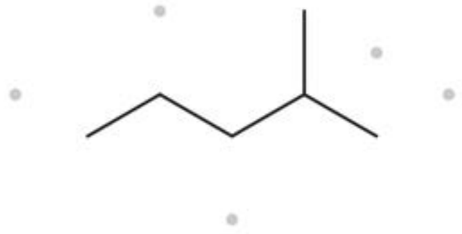


Parent: longest chain = 5 C ⁀ pentane.

Numbering: left ⁀ right gives {2}; right ⁀ left gives {4}. First point of difference: 2 < 4, so number from the left.

Substituents: methyl on C2 ⁀ 2-methyl.

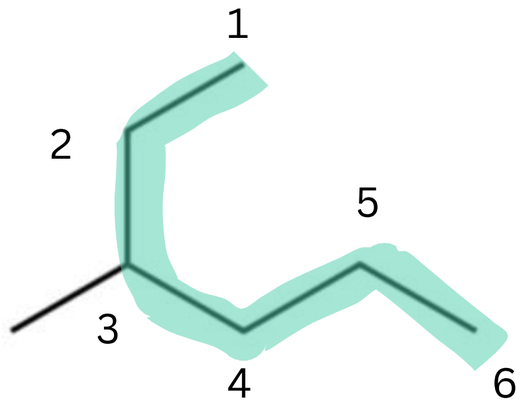
2-methylpentane



Returned Name: 2-methylpentane
Match ✓

✓ longest chain/ring ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

A02-FOUNDATION C_7H_{16}

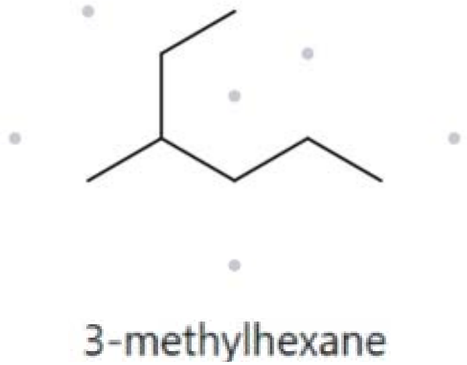


Parent: longest chain = 6 C ⁀ hexane (continuous chain is drawn bent; runs from top-left arm through branch down to bottom-right).

Numbering: top ⁀ bottom-right gives {3}; bottom-right ⁀ top gives {4}. First point of difference: 3 < 4, so number from top arm.

Substituents: methyl on C3 ⁀ 3-methyl.

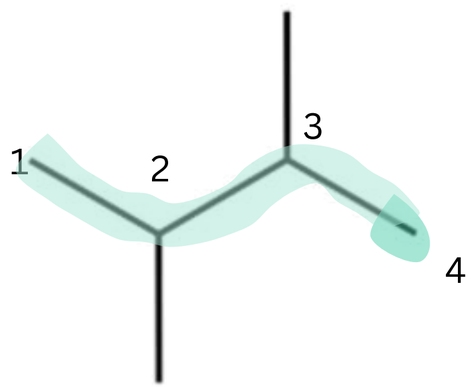
3-methylhexane



Returned Name: 3-methylhexane
Match ✓

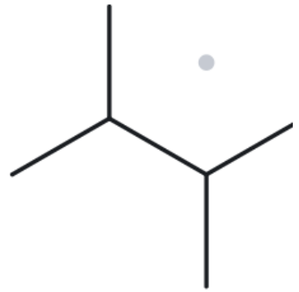
✓ longest chain/ring ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

A03-FOUNDATION C_6H_{14}



Parent: longest chain = 4 C ⁀ butane.
Numbering: both directions give 2,3, so either direction works.
Substituents (A⁀Z): two identical methyl groups on C2 and C3 ⁀ 2,3-dimethyl.

2,3-dimethylbutane

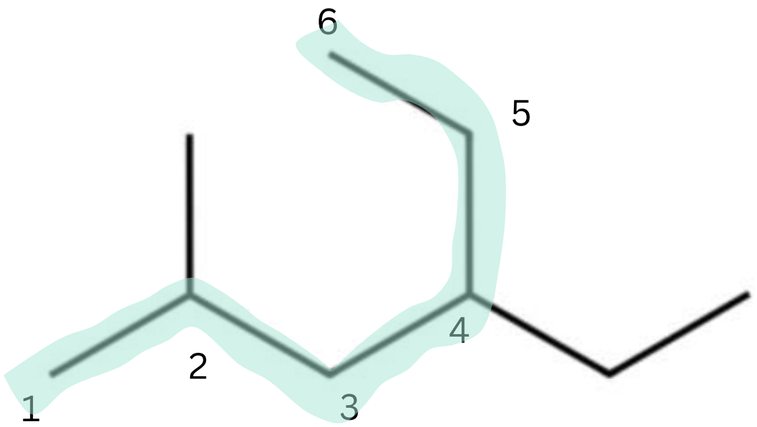


2,3-dimethylbutan

Returned Name: 2,3-dimethylbutane
Match ✓

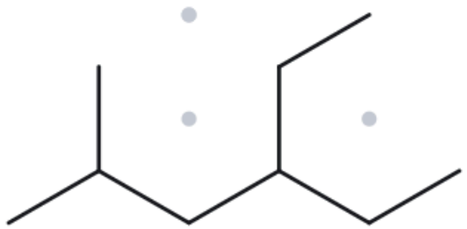
✓ longest chain/ring ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

A04-DEVELOPING C_9H_{20}



Parent: longest chain = 6 C ⁀ hexane. Two 6-C paths exist (through either two-carbon arm on C4), but both give the same name.
Numbering: lower left ⁀ upper right gives 2,4; the reverse gives 3,5. First point of difference: 2 < 3, so number from the lower left.
Substituents (A⁀Z): ethyl on C4, methyl on C2 ⁀ 4-ethyl-2-methyl (ethyl is cited first alphabetically).

4-ethyl-2-methylhexane

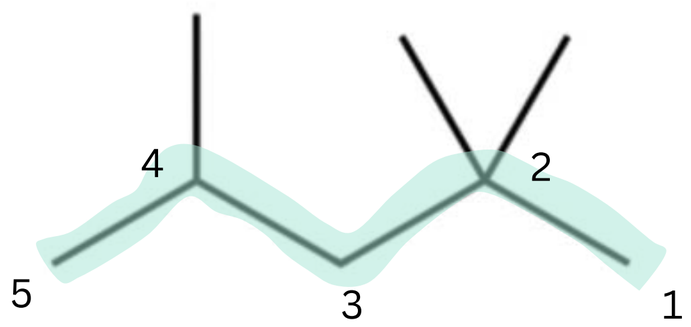


4-ethyl-2-methylhexane

Returned Name: 4-ethyl-2-methylhexane
Match ✓

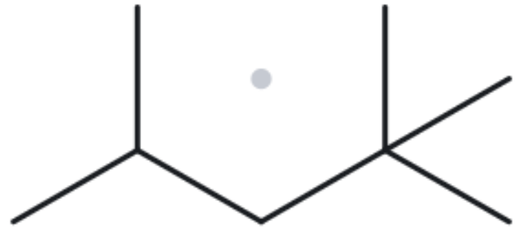
✓ longest chain/ring ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

A05-DEVELOPING C_8H_{18}



Parent: longest chain = 5 C ⁀ pentane.
Numbering: right ⁀ left gives 2,2,4; left ⁀ right gives 2,4,4. First point of difference: the second locant, 2 < 4, so number from the right.
Substituents: three methyl groups on C2, C2 and C4 ⁀ 2,2,4-trimethyl.

2,2,4-trimethylpentane

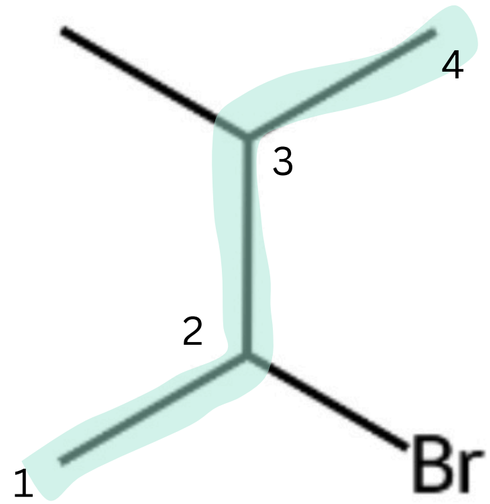


2,2,4-trimethylpentane

Returned Name: 2,2,4-trimethylpentane
Match ✓

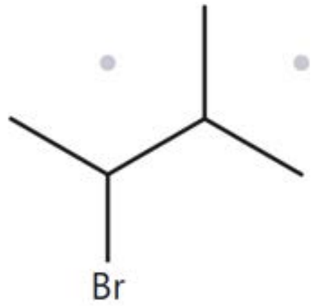
✓ longest chain/ring ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

A06-DEVELOPING $C_5H_{11}Br$



Parent: longest chain = 4 C ⁀ butane.
Numbering: bottom ⁀ top gives 2,3 (Br on C2, methyl on C3); top ⁀ bottom also gives 2,3. The locant sets tie, so the alphabetical tie-break applies: bromo comes before methyl, so Br gets the lower number.
Substituents (A⁀Z): bromo on C2, methyl on C3 ⁀ 2-bromo-3-methyl.

2-bromo-3-methylbutane

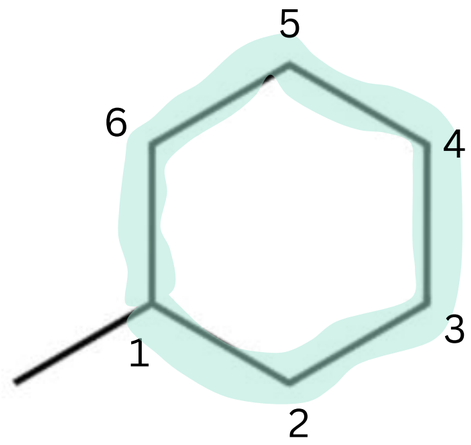


2-bromo-3-methylbutane

Returned Name: 2-bromo-3-methylbutane
Match ✓

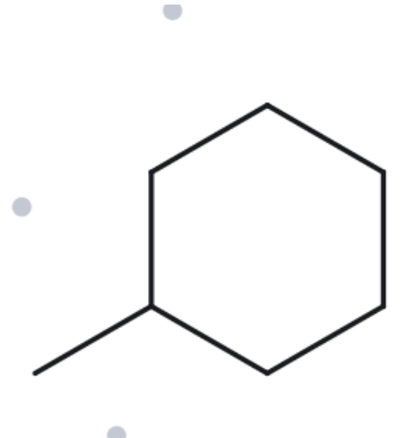
✓ longest chain/ring ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

A07-DEVELOPING C_7H_{14}



Parent: ring of 6 C ⁀ cyclohexane (the ring is the parent).
Numbering: with one substituent, the ring carbon bearing the methyl is C1 and the locant is not written.
Substituents: methyl ⁀ methylcyclohexane.

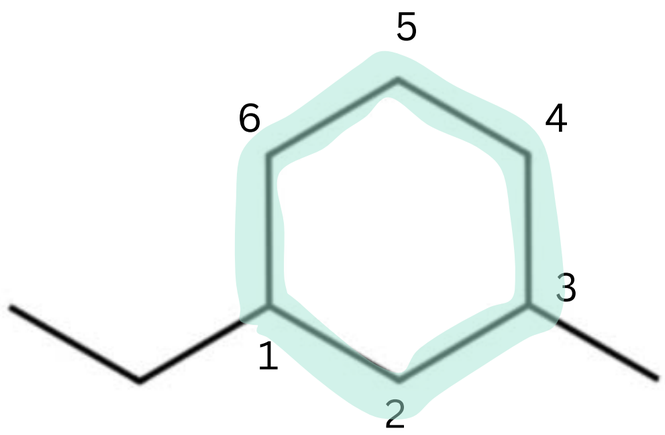
methylcyclohexane



Returned Name: methylcyclohexane
Match ✓

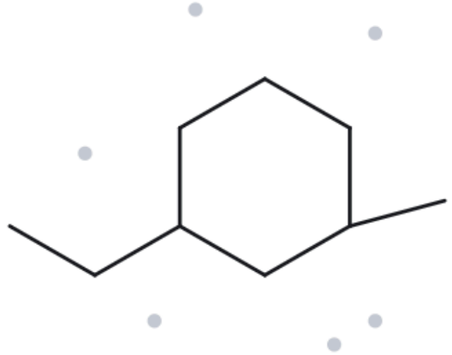
✓ longest chain/ring ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

A08-CHALLENGE C_9H_{18}



Parent: ring of 6 C ⁀ cyclohexane.
Numbering: either direction (the short way round) gives 1,3, so the locant sets tie.
Alphabetical tie-break: ethyl comes before methyl, so ethyl is C1.
Substituents (A⁀Z): ethyl on C1, methyl on C3 ⁀ 1-ethyl-3-methyl.

1-ethyl-3-methylcyclohexane

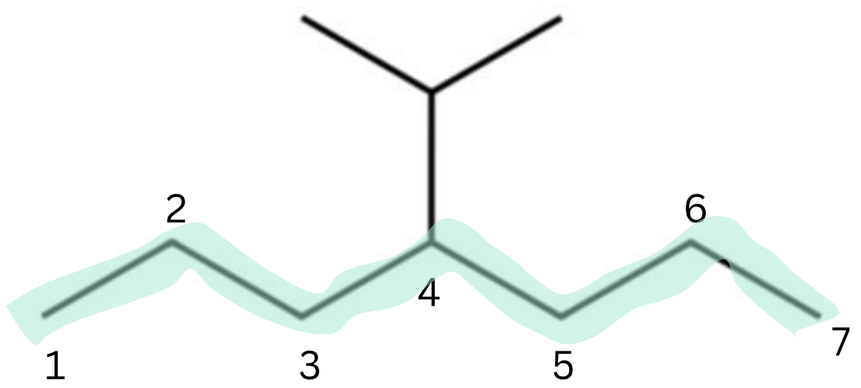


1-ethyl-3-methylcyclohexane

Returned Name: 1-ethyl-3-methylcyclohexane
Match ✓

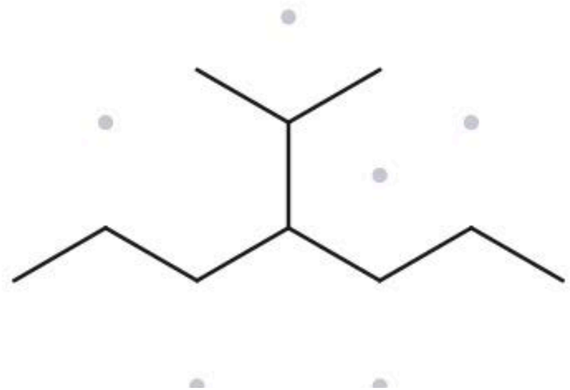
✓ longest chain/ring ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

A09-CHALLENGE $C_{10}H_{22}$



Parent: longest chain = 7 C, traced through the branch point (a path through the branch gives only 6 C) ⌒ heptane.
Numbering: either end gives 4, since the branch is on the central carbon.
Substituents: isopropyl, written (1-methylethyl), on C4 ⌒ 4-isopropyl.

4-isopropylheptane



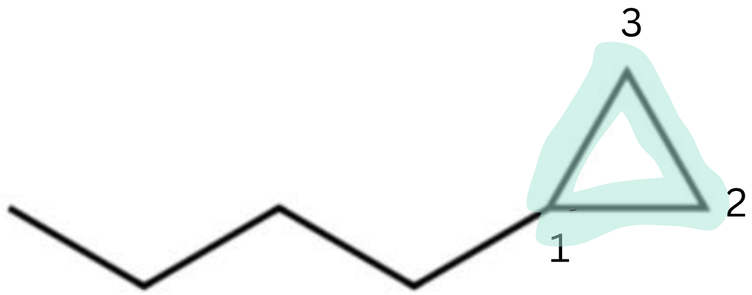
2-methyl-3-propylhexane

Correct name: 4-isopropylheptane; rule that decides: longest chain is 7 C, not the 6-C parent the tool gave.

Returned Name: 2-methyl-3-propylhexan
Match ✗

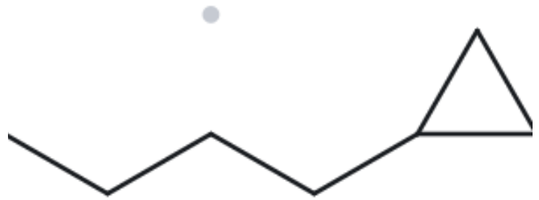
✓ longest chain/ring ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

A10-CHALLENGE C_7H_{14}



Parent: the ring is the parent in this activity (current IUPAC rule), even though the chain has 4 C and the ring 3 C ⁀ cyclopropane.
Numbering: C1 is the ring carbon bonded to the chain; with one substituent the locant is not written.
Substituents: butyl (the 4-carbon chain) ⁀ butylcyclopropane.

butylcyclopropane



butylcyclopropane

Returned Name: butylcyclopropane
Match ✓

✓ longest chain/ring ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

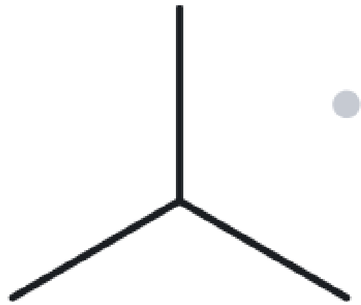
B01-FOUNDATION

Given Name: 2-methylpropane

NAME DECODED

Parent: Propane ⌒ Three (3) single-bonded C; chain
Substituents: methyl on C2
Expected formula: worked out from the name before drawing: C4H10

BOND-LINE STRUCTURE



FORMULA CHECK

C4H10
General formula: CnH2n+2 (open chain) = C4H10
Agrees? Yes

BACK-CHECK

Returned name: 2-methylpropane
Verdict: same as given

parent length ✓ substituent positions ✓ formula agrees ✓ back-check recorded ✓

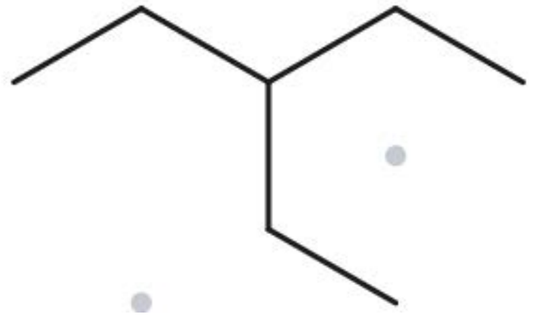
BO2-FOUNDATION

Given name: 3-ethylpentane

NAME DECODED

Parent: Pentane ⌒ Five (5) single-bonded C; chain
Substituents: Ethyl on C3
Expected formula: worked out from the name before drawing: C7H16

BOND-LINE STRUCTURE



FORMULA CHECK

C7H16
General formula: CnH2n+2 (open chain) = C7H16
Agrees? yes

BACK-CHECK

Returned name: 3-ethylpentane
Verdict: same as given

parent length ✓ substituent positions ✓ formula agrees ✓ back-check recorded ✓

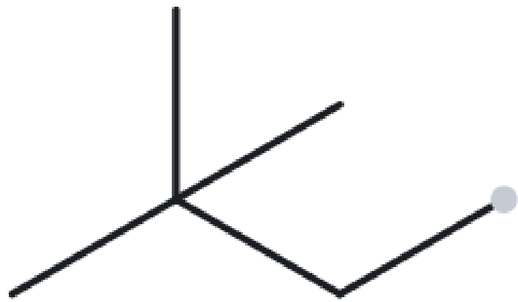
B03-FOUNDATION

Given name: 2,2-dimethylbutane

NAME DECODED

Parent: Butane ⁀ Four (4) single-bonded C; chain
Substituents: methyl on C2 ; methyl on C2
Expected formula: worked out from the name before drawing: C6H14

BOND-LINE STRUCTURE



FORMULA CHECK

C6H14
General formula: CnH2n+2 (open chain) = C6H14
Agrees? yes

BACK-CHECK

Returned name: 2,2-dimethylbutane
Verdict: same as given

parent length ✓ substituent positions ✓ formula agrees ✓ back-check recorded ✓

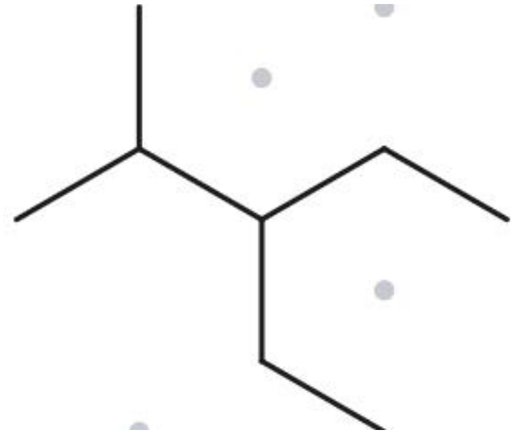
B04-DEVELOPING

Given name: 3-ethyl-2-methylpentane

NAME DECODED

Parent: Pentane ⁀ Five (5) single-bonded C; chain
Substituents: Ethyl on C3 ; methyl on C2
Expected formula: worked out from the name before drawing: C8H18

BOND-LINE STRUCTURE



FORMULA CHECK

C8H18
General formula: CnH2n+2 (open chain) = C8H18
Agrees? yes

BACK-CHECK

Returned name: 3-ethyl-2-methylpentane
Verdict: same as given

parent length ✓ substituent positions ✓ formula agrees ✓ back-check recorded ✓

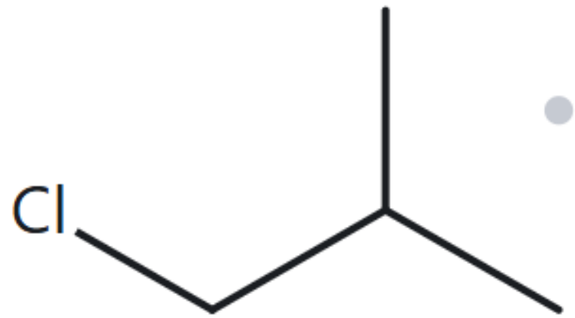
B05-DEVELOPING

Given name: 1-chloro-2-methylpropane

NAME DECODED

Parent: Propane ⌒ Three (3) single-bonded C; chain
Substituents: Chloro on C1 ; methyl on C2
Expected formula: worked out from the name before drawing: C4H9Cl

BOND-LINE STRUCTURE



FORMULA CHECK

C4H9Cl
General formula: CnH2n+2 (open chain) = C4H10; each halogen replaces one H = C4H9Cl
Agrees? yes

BACK-CHECK

Returned name: 1-chloro-2-methylpropane
Verdict: same as given

parent length ✓ substituent positions ✓ formula agrees ✓ back-check recorded ✓

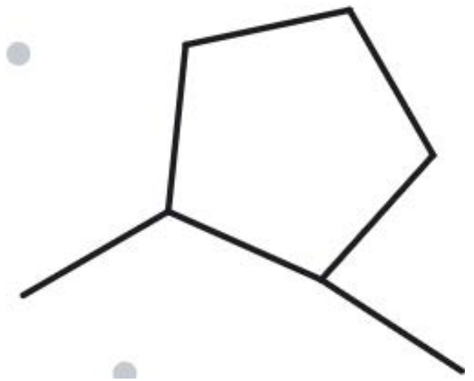
B06-DEVELOPING

Given name: 1,2-dimethylcyclopentane

NAME DECODED

Parent: Cyclopentane⁀ Five (5) single-bonded C; ring
Substituents: Methyl on C1 and C2
Expected formula: worked out from the name before drawing: C7H14

BOND-LINE STRUCTURE



FORMULA CHECK

C7H14
General formula: CnH2n (one ring) = C7H14
Agrees? yes

BACK-CHECK

Returned name: 1,2-dimethylcyclopentane
Verdict: same as given

parent length ✓ substituent positions ✓ formula agrees ✓ back-check recorded ✓

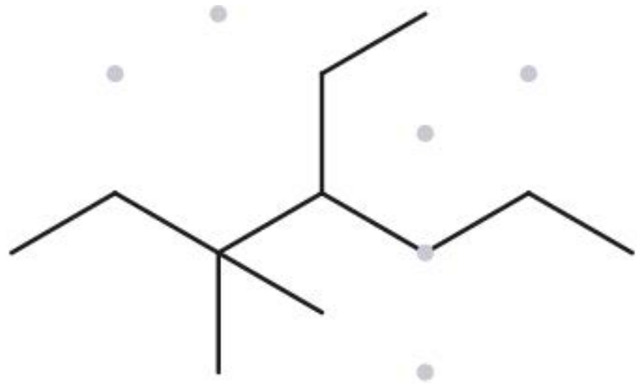
B07- DEVELOPING

Given name: 4-ethyl-3,3-dimethylheptane

NAME DECODED

Parent: Heptane ⁀ Seven (7) single-bonded C; chain
Substituents: Ethyl on C4 ; methyl on C3 (two)
Expected formula: worked out from the name before drawing: C11H24

BOND-LINE STRUCTURE



FORMULA CHECK

C11H24
General formula: CnH2n+2 (open chain) = C11H24
Agrees? yes

BACK-CHECK

Returned name: 4-ethyl-3,3-dimethylheptane
Verdict: same as given

parent length ✓ substituent positions ✓ formula agrees ✓ back-check recorded ✓

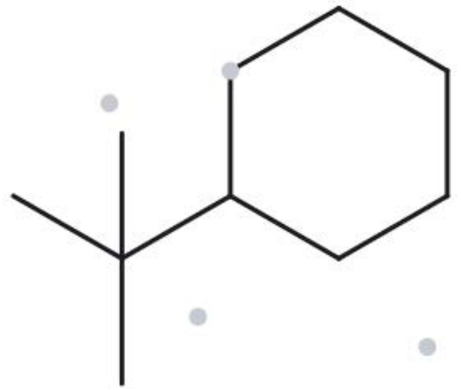
BOB-CHALLENGE

Given name: tert-butylcyclohexane

NAME DECODED

Parent: Cyclohexane⁀ Six (6) single-bonded C; ring
Substituents: tert-butyl on C1
Expected formula: worked out from the name before drawing: C10H20

BOND-LINE STRUCTURE



FORMULA CHECK

C10H20
General formula: CnH2n (one ring) = C10H20
Agrees? yes

BACK-CHECK

Returned name: (1,1-dimethylethyl)cyclohexane
Verdict: differs
Explanation: same compound written with a different accepted substituent name. No drawing error.

parent length ✓ substituent positions ✓ formula agrees ✓ back-check recorded ✓

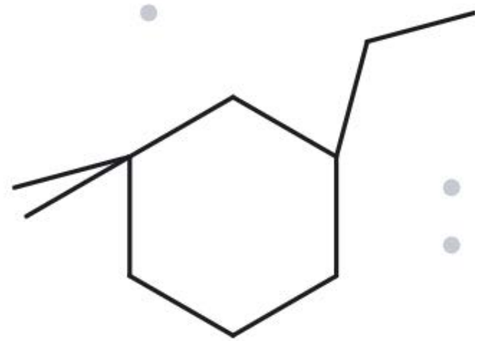
BOO-CHALLENGE

Given name: 3-ethyl-1,1-dimethylcyclohexane

NAME DECODED

Parent: Cyclohexane ⁀ Six (6) single-bonded C; ring
Substituents: Ethyl on C3 ; dimethyl on C1
Expected formula: worked out from the name before drawing: C10H20

BOND-LINE STRUCTURE



FORMULA CHECK

C10H20
General formula: CnH2n (one ring) = C10H20
Agrees? yes

BACK-CHECK

Returned name: 3-ethyl-1,1-dimethylcyclohexane
Verdict: same as given

parent length ✓ substituent positions ✓ formula agrees ✓ back-check recorded ✓

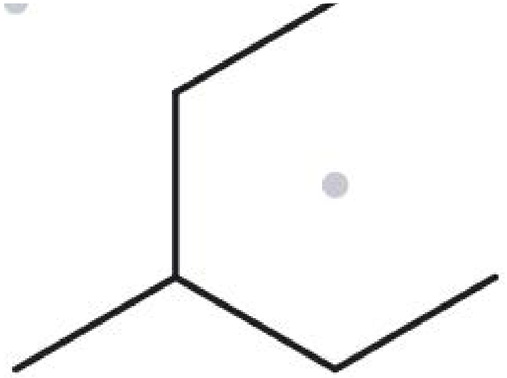
B1Q-CHALLENGE

Given name: 2-ethylbutane

NAME DECODED

Parent: Butane ⁀ Four (4) single-bonded C; chain
Substituents: Ethyl on C2
Expected formula: worked out from the name before drawing: C6H14

BOND-LINE STRUCTURE



FORMULA CHECK

C6H14
General formula: CnH2n+2 (open chain) = C6H14
Agrees? yes

BACK-CHECK

Returned name: 3-methylpentane
Verdict: same as given

Explanation: This given name incorrect.
Correct name: 3-methylpentane;
rule broken: the parent chain must be the longest continuous chain.

parent length ✓ substituent positions ✓ formula agrees ✓ back-check recorded ✓